YEAR 10 SCIENCE (GENERAL)
PHYSICAL SCIENCE
TEST 1

Name: ____________________________________ **Mark:** $\frac{\quad}{18}$

Part 1: Multiple Choice

Write your choice for each question in the table below.

MULTIPLE CHOICE ANSWERS	
1	
2	
3	
4	
5	
6	

1. How long would a car travelling at a speed of 20 m/s take to cover a distance of 360 m?

   (a) 13 s
   (b) 26 s
   (c) 18 s
   (d) 34 s

2. Speed is a measure of the rate:

   (a) at which distance is covered.
   (b) at which speed increases.
   (c) of change in force.
   (d) of change in acceleration.

3. Belinda walks a total distance of 1200 metres to school each morning. If this trip takes her 800 seconds, her average speed would be:

   (a) 1.5 m/s.
   (b) 0.67 m/s.
   (c) 0 m/s.
   (d) 960,000 m/s.

4. A car is driven along a highway at an average speed of 90 km/h for 2 hours without a rest stop. What distance has it covered?

   (a) 45 km
   (b) 180 km
   (c) 360 km
   (d) Not enough information is given to calculate.

5. A car travels 400 km on a highway at a speed of 80 km/h. What time does the trip take?

   (a) 0.5 hours
   (b) 5 hours
   (c) 20 hours
   (d) 50 hours

6. The sample of ticker tape below shows:

(a) a uniformly accelerating vehicle.
(b) a vehicle travelling at a constant velocity.
(c) a stationary vehicle.
(d) a uniformly decelerating vehicle.

Part 2: Short Answer

1. Describe or define the following terms. (2 marks)

(a) distance:

(b) displacement:

2. Decide whether each of the following statements is **true (T)** or **false (F)**. (3 marks)

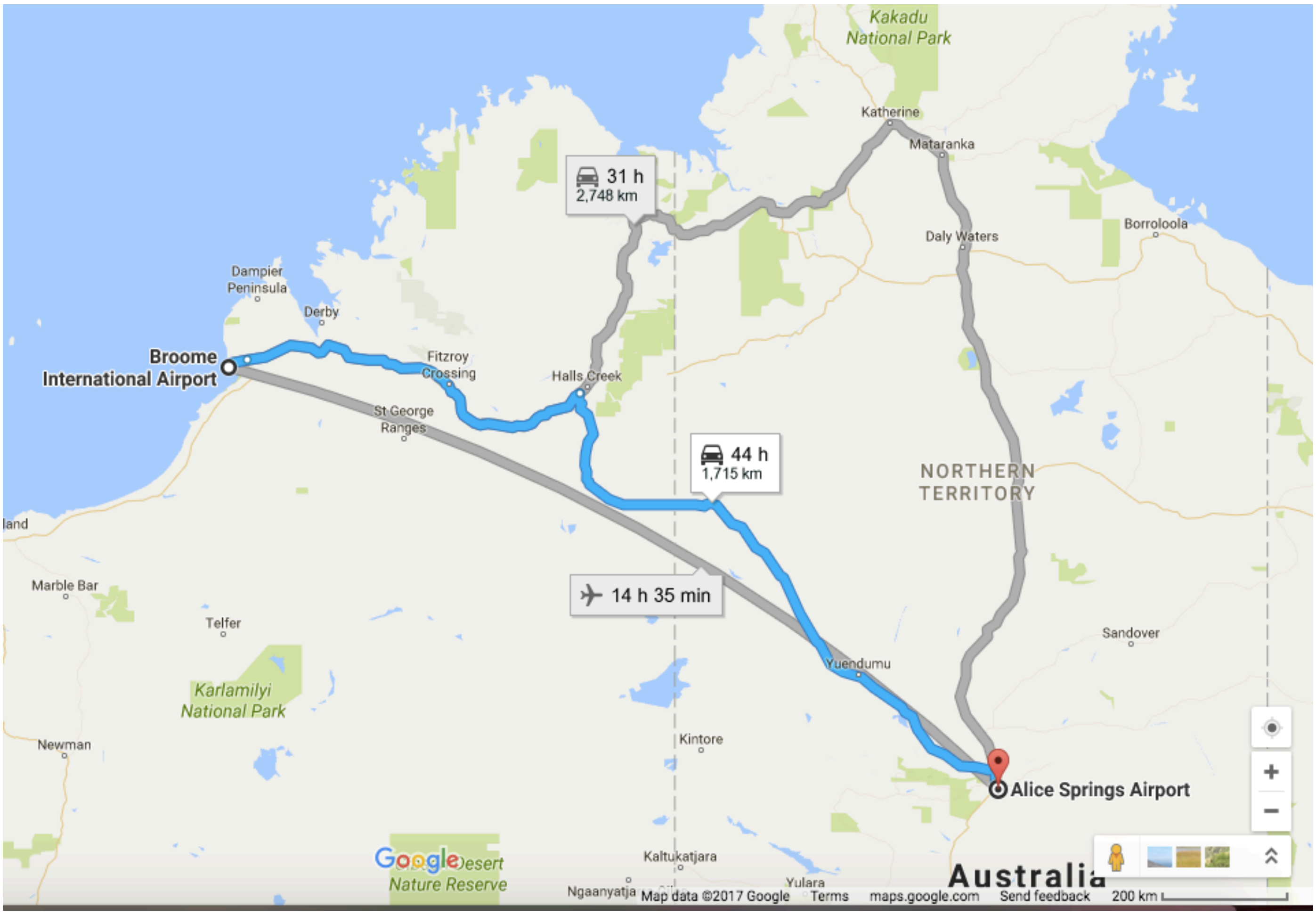
- Average speed is calculated by distance covered multiplied by the time taken. _____
- 40.0 km/h = 144 m/s _____
- The relationship between time, distance and speed is $t = \frac{sp}{d}$. _____

3. The map below shows Broome and Alice Springs. Use the map and its scale to work out the following.

(a) The ***distance*** between Broome Airport and Alice Springs airport. (1 mark)

(b) The ***displacement*** between the two airports. (**HINT**: Don't forget to include a ***bearing*** and ***start from Broome***.)

(3 marks)



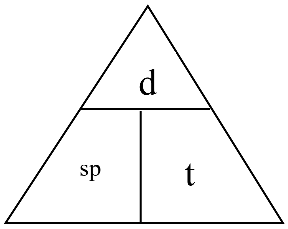
4. Complete the following table on ***speed*** calculations. Make sure you include the appropriate units for each quantity.

(3 marks)

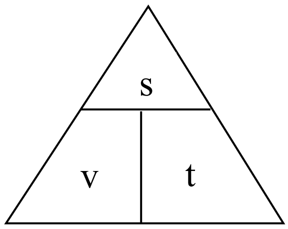
Distance travelled	Time taken	Average speed
490 m	35 s	
	12 s	23 m/s
121 m		11 m/s

YEAR 10 SCIENCE
PHYSICAL SCIENCES
PHYSICS DATA SHEET

Speed $\text{speed} = \dfrac{d}{t}$

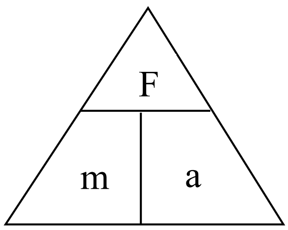


Velocity $v = \dfrac{s}{t}$

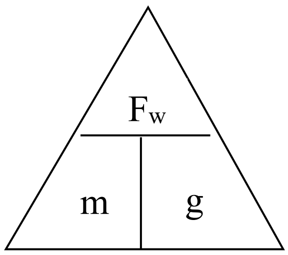


Acceleration $a = \dfrac{v - u}{t}$ $v = u + at$

Force $F = ma$



Weight $F_w = mg$



Acceleration due to gravity $g = 9.80\ ms^{-2}$

Converting km/h to m/s ***Divide*** the speed in km/h by 3.6 to get the speed in m/s.